

1.1 Physical Quantities & Units

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Hard

Time allowed: 60

Score: /48

Percentage: /100

Question 1a

Convert the following measurements

(i)
42 km³ into mm³ [2]

(ii)
25 light-years into nm [3]

(iii)
0.05 kWh into PeV [3]

[8 marks]

Question 1b

The frequency f of a stationary wave on a string is given by

$$f^k = \frac{T}{4mL^2}$$

where L is the length of the string, T is the tension in the string, m is the mass per unit length of the string and k is a dimensionless constant.

By considering the homogeneity of the equation, determine the value of k .

[3 marks]

Question 1c

The current I through a semiconductor diode given by

$$I = I_0 e^{-\frac{qV}{nT}}$$

where I_0 is the initial current, q is the charge of an electron, V is the applied potential difference, T is the temperature and n is a constant.

(i)

By considering the homogeneity of the equation, determine the units of n .

[3]

(ii)

Use the datasheet to determine the name of this constant.

[1]

[4 marks]

Question 1d

Complete the following table by giving the SI base units. Then estimate the order of magnitude for each of the physical quantities.

Physical Quantity	SI Base Unit	Order of Magnitude
Acceleration of freefall Earth (g)	m s^{-2}	
Stephan-Boltzmann constant (σ)		10^{-7}
Speed of a β particle		
Specific heat capacity of water (c)		

[4 marks]**Question 2a**

Estimate the time it takes light to cross the nucleus of a hydrogen atom.

[2 marks]

Question 2b

A foil made from gold has a thickness of $0.132\text{ }\mu\text{m}$. A gold atom has a radius of 146 pm .

(i)

Estimate the number of layers of atoms in the gold foil to the nearest hundred.

[2]

(ii)

The radius of a gold nucleus is about 20 000 times smaller than its atomic radius. The cross-sectional area of a nucleus is commonly measured using the unit *barn*, which is represented by the symbol *b*.

Estimate the cross-sectional area of a gold nucleus in barns (b).

$$1\text{ b} = 100\text{ fm}^2$$

[4]

[6 marks]

Question 2c

The atomic mass unit *u* is a common unit used in nuclear physics. $1u$ is equivalent to 931.5 MeV c^{-2}

Show that $1u$ is approximately equal to the mass of a proton in kg.

[3 marks]

Question 2d

An acre is a unit of land measurement approximately equal to 4050 m^2 which is roughly equivalent to the size of a football field.

An Astronomical Unit, or 1 AU, is defined as the average distance between the Earth and the Sun and is approximately equal to 150 million km.

Estimate the number of football fields that could fit lengthways between the Earth and the Sun.

[3 marks]

Question 3a

Fig. 1.1 shows two taut, light ropes that keep a pole vertically upright by applying two tension forces, one of magnitude 200 N and one of magnitude T .

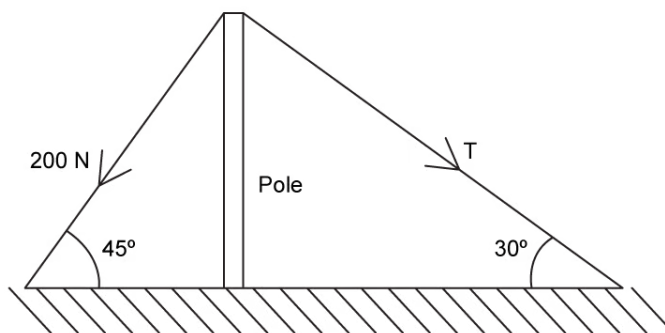


Fig. 1.1

By constructing a scale diagram, determine the weight of the pole W and the magnitude of T .

[4 marks]

Question 3b

A canoeist can paddle at a speed of 3.8 m s^{-1} in still water. The canoeist encounters an opposing current, moving at a speed of 1.5 m s^{-1} at 30° to their original direction of travel, as shown in Fig. 1.2.

**Fig. 1.2**

Using Fig. 1.2 to construct a scale diagram, determine the magnitude of the canoeist's resultant velocity.

[3 marks]

Question 3c

The boat shown in Fig. 1.3 is being towed at a constant velocity by a towing rope, which exerts a tension force $F_T = 2500 \text{ N}$.

There are two resistive forces indicated; the force of the water on the keel F_K and the force of the water on the rudder, F_R .

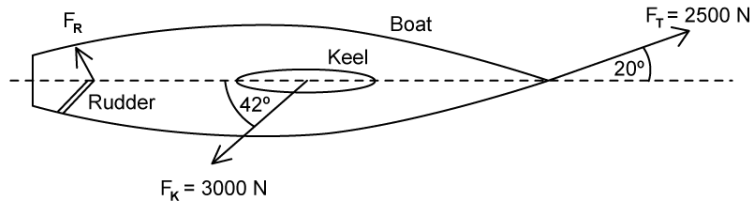


Fig. 1.3

By scale diagram or otherwise, determine the magnitude of F_R .

[4 marks]

Question 3d

Another boat wishes to cross a river. The river flows from west to east at a constant velocity of 35 cm s^{-1} and the boat leaves the south bank, due north, at 1.5 m s^{-1} as shown in Fig. 1.4.

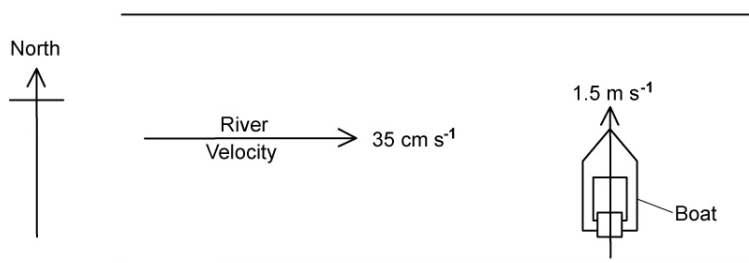


Fig. 1.4

Construct a scale diagram to show that the new direction of the boat is about 13° East of North.

[4 marks]

Model Answers: Hard

1a

(a)

(i) Convert 42 km^3 into mm^3 :

- Convert km into mm:

$$1 \text{ km} = \frac{1 \times 10^3}{1 \times 10^{-3}} = 1 \times 10^6 \text{ mm}$$

- Convert km^3 into mm^3 :

$$1 \text{ km}^3 = (1 \times 10^6)^3 \text{ mm}^3 = 1 \times 10^{18} \text{ mm}^3 \text{ [1 mark]}$$

$$42 \text{ km}^3 = 42 \times (1 \times 10^{18}) = 42 \times 10^{18} \text{ mm}^3 \text{ [1 mark]}$$

(ii) Convert 25 light-years into nm:

- 1 light year = the distance light travels in 1 year
- Speed of light = $3.0 \times 10^8 \text{ m s}^{-1}$

- Convert 1 year into s:

$$1 \text{ year} = 60 \times 60 \times 24 \times 365 = 3.15 \times 10^7 \text{ s}$$

- Convert 1 ly into m:

$$\text{distance} = \text{speed} \times \text{time}$$

$$1 \text{ ly} = (3.0 \times 10^8) \times (3.15 \times 10^7) = 9.46 \times 10^{15} \text{ m [1 mark]}$$

- Convert m into nm:

$$1 \text{ ly} = \frac{9.46 \times 10^{15}}{1 \times 10^{-9}} = 9.46 \times 10^{24} \text{ nm [1 mark]}$$

- $25 \text{ ly} = 25 \times (9.46 \times 10^{24}) = 2.4 \times 10^{26} \text{ nm (2 s.f.) [1 mark]}$

(iii) Convert 0.05 kWh into PeV:

- Convert from kWh to Wh:
 $1 \text{ kW} = 1000 \text{ W}$
 $0.05 \text{ kWh} = 0.05 \times 1000 = 50 \text{ Wh}$
- Convert from Wh to J:
 $1 \text{ h} = 60 \times 60 = 3600 \text{ s}$
 $50 \text{ Wh} = 50 \times 3600 = 180\,000 \text{ W s} = 180\,000 \text{ J}$ [1 mark]
- Convert J to eV:
 $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
 $180\,000 \text{ J} = \frac{180\,000}{1.60 \times 10^{-19}} = 1.125 \times 10^{24} \text{ eV}$ [1 mark]
- Convert from eV to PeV:
 $1 \text{ PeV} = 1 \times 10^{15} \text{ eV}$
- $1.125 \times 10^{24} \text{ eV} = \frac{1.125 \times 10^{24}}{1 \times 10^{15}} = 1.1 \times 10^9 \text{ PeV}$ [1 mark]

[Total: 8 marks]

1b

(b) By considering the homogeneity of the equation, determine the value of k :

- $f^k = \frac{T}{4mL^2}$
- Consider the units on the LHS: $(\text{Hz})^k = \left(\frac{1}{\text{s}}\right)^k$ [1 mark]
- Consider the units on the RHS: $\frac{\text{N}}{(\text{kg m}^{-1}) \text{ m}^2}$ [1 mark]

Simplify the units on the RHS:

- From $F = ma$, the SI base unit of N is kg m s^{-2}

- Hence $\frac{N}{(kg\ m^{-1})\ m^2} = \frac{kg\ m\ s^{-2}}{kg\ m} = s^{-2}$
- Comparing RHS with LHS: $s^{-k} = s^{-2}$
- Therefore, $k = 2$ [1 mark]

[Total: 3 marks]

1c

(c)

(i) By considering the homogeneity of the equation, determine the units of n :

- $I = I_0 e^{-\frac{qV}{nT}} \Rightarrow \frac{I}{I_0} = e^{-\frac{qV}{nT}}$
- Since the units on the LHS $\frac{A}{A} = 1$, the units of $\frac{qV}{nT}$ will also be dimensionless

State the SI units of temperature T :

- Units of $[T] = K$

Determine the SI units of charge q :

- Charge: $I = \frac{q}{t} \Rightarrow q = It$
- Units of $[q] = C = A\ s$ [1 mark]

Determine the SI units of potential difference V :

- Potential difference: $V = \frac{W}{q}$

- Work done: $W = F \times s$ or $\frac{1}{2}mv^2$
- Units of $[W] = [m][v]^2 = J = \text{kg m}^2 \text{s}^{-2}$
- Units of $[V] = J C^{-1} = J A^{-1} \text{s}^{-1}$
- Units of $[V] = (\text{kg m}^2 \text{s}^{-2}) \text{s}^{-1} A^{-1} = \text{kg m}^2 \text{s}^{-3} A^{-1}$ [1 mark]

Determine the units of constant n :

- Units of n : $[n] = \frac{[q][V]}{[T]}$
- $[n] = \frac{A \text{ s} (\text{kg m}^2 \text{s}^{-3} A^{-1})}{K}$
- $[n] = \text{kg m}^2 \text{s}^{-2} K^{-1}$ [1 mark]

(ii) Determine the name of the constant:

- Since $J = \text{kg m}^2 \text{s}^{-2}$, the derived unit of n is $J K^{-1}$
- Therefore, this is the Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ [1 mark]

[Total: 4 marks]

1d

(d) The correct SI base units and estimates for order of magnitude are:

One mark for each correct row:

Quantity	SI Base Unit	Order of Magnitude
Acceleration of freefall Earth (g)	m s^{-2}	10^1
Stephan-Boltzmann constant (σ)	$\text{kg s}^{-3} \text{K}^{-4}$	10^{-7}
Speed of a β particle	m s^{-1}	10^8
Specific heat capacity of water (c)	$\text{m}^2 \text{s}^{-2} \text{K}^{-1}$	10^3

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[Total: 4 marks]

The acceleration of freefall on Earth, g , is 9.81 m s^{-2} . This quantity is already given in SI base units and the order of magnitude is 10^1 as $10^1 = 10$.

The Stephan-Boltzmann constant σ is given in the data booklet as $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Rounded to the nearest power of ten, the order of magnitude is 10^{-7} .

The units of the Stephan-Boltzmann constant σ are:

Watts (W) are the unit for power

Power = $\frac{\text{Energy}}{\text{time}}$

$W = J s^{-1}$

$W m^{-2} K^{-4}$

$J s^{-1} m^{-2} K^{-4}$

$J = kg m^2 s^{-2}$

$kg m^2 s^{-2} s^{-1} m^{-2} K^{-4}$

$kg m^2 s^{-2} s^{-1} m^{-2} K^{-4}$

$m^2 m^{-2}$ cancel

$kg s^{-2} s^{-1} K^{-4}$

$s^{-1} s^{-2} = s^{-3}$

$kg s^{-3} K^{-4}$

The speed of a β particle can be approximated to the speed of light since a β particle is a high-speed electron. The speed of light in a vacuum c is given in the data booklet as $3.00 \times 10^8 \text{ m s}^{-1}$

Metres and seconds are already SI base units, and rounded to the nearest power of 10, the order of magnitude is 10^8

The specific heat capacity c of water is not given in the data booklet, but you should be familiar with it from practising calculations. The specific heat capacity of water is about $4200 \text{ J kg}^{-1} \text{ K}^{-1}$, or $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Rounded to the nearest power of 10, the order of magnitude is 10^3 and the units are:

$$\begin{array}{l}
 \text{J kg}^{-1} \text{ K}^{-1} \\
 \swarrow \\
 \text{J} = \text{kg m}^2 \text{ s}^{-2} \\
 \searrow \\
 \text{kg m}^2 \text{ s}^{-2} \text{ kg}^{-1} \text{ K}^{-1} \\
 \text{kg m}^2 \text{ s}^{-2} \text{ kg}^{-1} \text{ K}^{-1} \\
 \text{kg kg}^{-1} \text{ cancels} \\
 \text{m}^2 \text{ s}^{-2} \text{ K}^{-1}
 \end{array}$$

2a

(a) Estimate the time it takes light to cross the nucleus of a hydrogen atom:

- Speed of light in vacuum, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Estimate the diameter of the nucleus of a hydrogen atom:

- A hydrogen atom has a nucleus made of 1 proton
- The diameter of a proton is roughly $1 \text{ fm} = 1 \times 10^{-15} \text{ m}$ [1 mark]

Estimate the time using the equation relating speed, time and distance:

- $\text{time} = \frac{\text{distance}}{\text{speed}}$
- $\text{time} = \frac{1 \times 10^{-15}}{3.00 \times 10^8} = 3 \times 10^{-24} \text{ s (1 s.f.)}$ [1 mark]

[Total: 2 marks]

2b

(b)

List the known quantities:

- Thickness of gold foil = $0.132 \mu\text{m} = 0.132 \times 10^{-6} \text{ m}$
- Radius of gold atom = $146 \text{ pm} = 146 \times 10^{-12} \text{ m}$
- Radius of gold nucleus = $\frac{\text{radius of gold atom}}{20\,000}$

(i) Estimate the number of layers of atoms in the gold foil to the nearest hundred:

- Diameter of a gold atom = $(146 \times 10^{-12}) \times 2 = 292 \times 10^{-12} \text{ m}$ [1 mark]
- Number of layers = $\frac{0.132 \times 10^{-6}}{292 \times 10^{-12}} = 452.05 = 500 \text{ layers (1 s.f.)}$ [1 mark]

(ii) Calculate the cross-sectional area of a gold nucleus in barns (b):

- Convert fm^2 into m^2 :
 $1 \text{ fm} = 1 \times 10^{-15} \text{ m}$
 $1 \text{ fm}^2 = (1 \times 10^{-15})^2 \text{ m}^2 = 1 \times 10^{-30} \text{ m}^2$ [1 mark]
- Calculate the value of 1 barn in m^2 :
 $1 \text{ b} = 100 \text{ fm}^2 = 100 \times (1 \times 10^{-30}) \text{ m}^2 = 1 \times 10^{-28} \text{ m}^2$ [1 mark]

Determine the cross-sectional area and convert to barns:

- Radius of a gold nucleus: $r \approx \frac{146 \times 10^{-12}}{20\,000} \approx 7.3 \times 10^{-15} \text{ m}$
- Cross-sectional area of a gold nucleus = $\pi r^2 \approx \pi(7.3 \times 10^{-15})^2 \approx 1.67 \times 10^{-28} \text{ m}^2$ [1 mark]
- Cross-sectional area of a gold nucleus = $\frac{1.67 \times 10^{-28}}{1 \times 10^{-28}} \approx 1.67 \text{ b}$ [1 mark]

[Total: 6 marks]

Although the values in the question are given to three significant figures, it makes sense to round to the nearest hundred in part (i), since we are ignoring factors such as spaces between atoms and layers in the lattice.

The barn described in this question is a real unit. Other less well-known units include the

Although the values in the question are given to three significant figures, it makes sense to round to the nearest hundred in part (i), since we are ignoring factors such as spaces between atoms and layers in the lattice.

The barn described in this question is a real unit. Other less well-known units include the 'outhouse' ($1 \mu\text{b}$) and the 'shed' (10^{-24} b). Particle physicists are clearly quite creative!

2c

(c) Show that $1u$ is approximately equal to the mass of a proton in kg:

List the known quantities:

- Atomic mass unit, $1 u = 931.5 \text{ MeV } c^{-2}$
- Speed of light in vacuum, $c = 3.00 \times 10^8 \text{ m s}^{-1}$
- Mass of a proton, $m_p = 1.67 \times 10^{-27} \text{ kg}$

Convert $1u$ to J:

- $1u = 931.5 \text{ MeV} = 931.5 \times 10^6 \text{ eV}$ [1 mark]
- $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
- $1u = (931.5 \times 10^6) \times (1.60 \times 10^{-19}) = 1.4904 \times 10^{-10} \text{ J}$ [1 mark]

Convert energy in J to mass in kg:

- Energy-mass equivalence: $\Delta E = \Delta mc^2 \Rightarrow \Delta m = \frac{\Delta E}{c^2}$
- $m = \frac{1.4904 \times 10^{-10}}{(3.00 \times 10^8)^2} = 1.66 \times 10^{-27} \text{ kg}$ [1 mark]

$\therefore 1u$ is approximately equal to the mass of a proton $m_p = 1.67 \times 10^{-27} \text{ kg}$

[Total: 3 marks]

You will be expected to remember the definition of an electronvolt $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$ for your exam, however, you will be given the charge on an electron e on your data sheet which is also equal to $1.60 \times 10^{-19} \text{ C}$

2d

(d) Estimate the number of football fields that could fit lengthways between the Earth

and the Sun:

- $1 \text{ AU} = 150 \text{ million km} = 150 \times 10^6 \times 10^3 \text{ m} = 150 \times 10^9 \text{ m}$ [1 mark]
- $1 \text{ acre} = 1 \text{ football field} = 4050 \text{ m}^2$

Estimate the length of a football field

- A football field is rectangular, so a reasonable ratio of the length to the width of a football field might be 3:2
- This means area: $A = L \times w \Rightarrow A = 3x \times 2x$
- Where x = the constant of proportionality in the 3:2 ratio, i.e. $L = 3x$ and $w = 2x$
- $4050 = 6x^2 \Rightarrow x = \sqrt{\frac{4050}{6}} = 25.981 \approx 26 \text{ m}$
- Hence, length: $L = 3x \approx 3 \times 26 \approx 78 \text{ m}$ [1 mark]

Estimate the number of lengths that would fit in 1 AU:

- Number of football fields in 1 AU = $\frac{150 \times 10^9}{78} = 1.9231 \times 10^9 \approx 2 \times 10^9$ (1 s.f.) [1 mark]

$\therefore$ Approximately 2 billion football fields could fit lengthways between the Earth and the Sun

[Total: 3 marks]

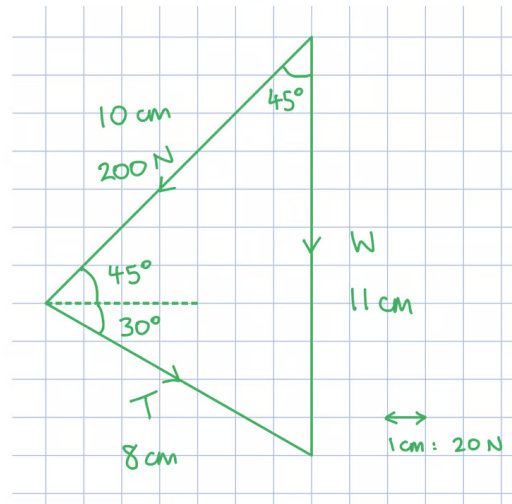
Questions such as this are all about your problem-solving skills. While approximating a football field to a square might not be exactly right, if you took this approach then you would still end up with the same answer once you round down to 1 significant figure

Approximating the field as a square: length = $\sqrt{4050} \approx 64 \text{ m}$

Number of football fields in 1 AU = $\frac{150 \times 10^9}{64} = 2.3438 \times 10^9 \approx 2 \times 10^9$ (1 s.f.)

3a

(a) Determine the magnitudes of W and T :



- Scale indicated as 1 cm : 20 N (or any appropriate scale); [1 mark]
- Correct resultant direction for weight W (vertically downwards, as shown); [1 mark]

Measurement of weight W :

- For a scale of 1 cm : 20 N
- Acceptable range for measurement: 11 – 11.5 cm
- Acceptable range for weight: 220 – 230 N [1 mark]

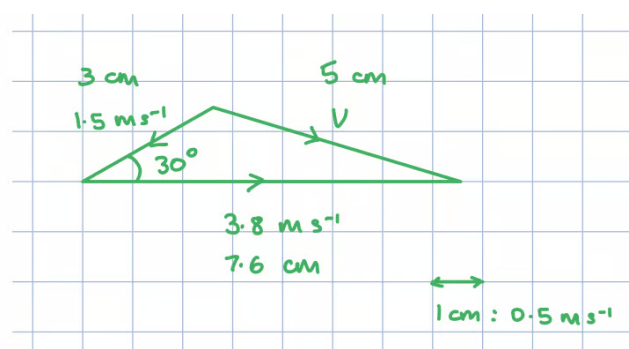
Measurement of tension, T :

- For a scale of 1 cm : 20 N
- Acceptable range for measurement: 8 – 8.5 cm
- Acceptable range for weight: 160 – 170 N [1 mark]

[Total: 4 marks]

3b

(b) Determine the canoeist's resultant velocity:



- Scale indicated as $1 \text{ cm} : 0.5 \text{ m s}^{-1}$ [1 mark]
- Correct resultant velocity indicated, with direction (as shown); [1 mark]

Measurement of resultant velocity:

- For a scale of $1 \text{ cm} : 0.5 \text{ m s}^{-1}$
- Acceptable range for measurement: $5.0 - 5.4 \text{ cm}$
- Acceptable range for velocity: $2.5 - 2.7 \text{ m s}^{-1}$ [1 mark]

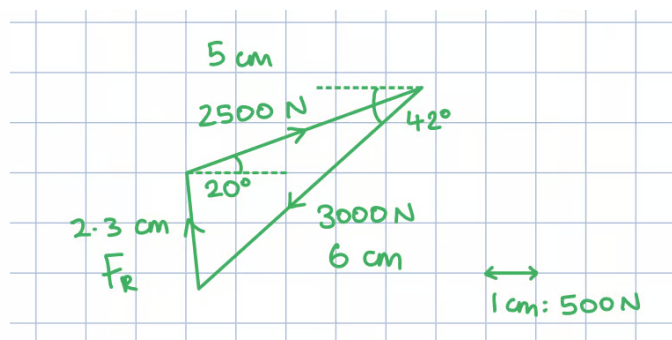
[Total: 3 marks]

3c

(c) Determine the magnitude of F_R :

List the known quantities:

- Tension in rope, $F_T = 2500 \text{ N}$
- Force on keel, $F_K = 3000 \text{ N}$



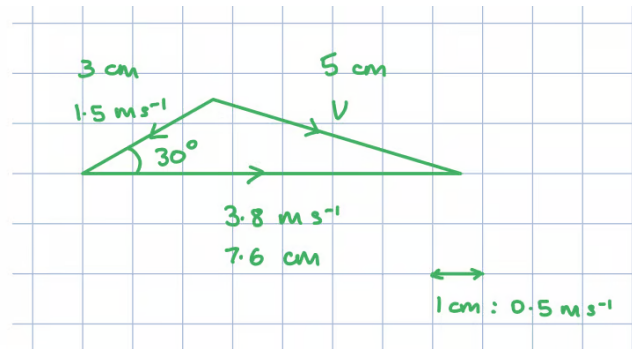
- Scale indicated as $1 \text{ cm} : 500 \text{ N}$ (or any appropriate scale); [1 mark]
- Correct directions and labels (as shown); [1 mark]

Measurement of the force F_R :

- For a scale of $1 \text{ cm} : 500 \text{ N}$
- Acceptable range for measurement: $2.2 - 2.5 \text{ cm}$ [1 mark]
- Acceptable range for F_R : $1100 - 1250 \text{ N}$ [1 mark]

[Total: 4 marks]

Note that the boat in this question is moving at a **constant velocity**: therefore, your vector triangle must be **closed** (i.e., the vectors begin and end at the same point, when



- Scale indicated as $1 \text{ cm} : 0.5 \text{ m s}^{-1}$ [1 mark]
- Correct resultant velocity indicated, with direction (as shown); [1 mark]

Measurement of resultant velocity:

- For a scale of $1 \text{ cm} : 0.5 \text{ m s}^{-1}$
- Acceptable range for measurement: $5.0 - 5.4 \text{ cm}$
- Acceptable range for velocity: $2.5 - 2.7 \text{ m s}^{-1}$ [1 mark]

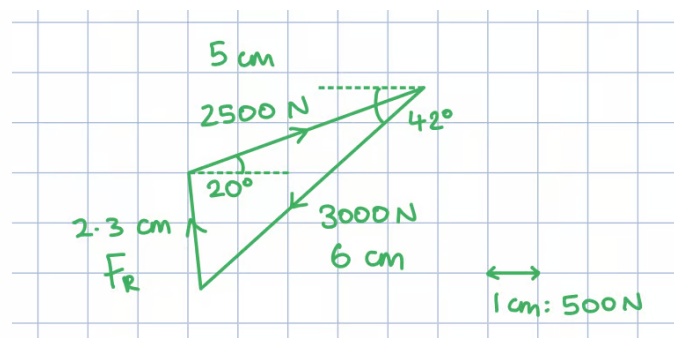
[Total: 3 marks]

3c

(c) Determine the magnitude of F_R :

List the known quantities:

- Tension in rope, $F_T = 2500 \text{ N}$
- Force on keel, $F_K = 3000 \text{ N}$



- Scale indicated as $1 \text{ cm} : 500 \text{ N}$ (or any appropriate scale); [1 mark]
- Correct directions and labels (as shown); [1 mark]

Measurement of the force F_R :

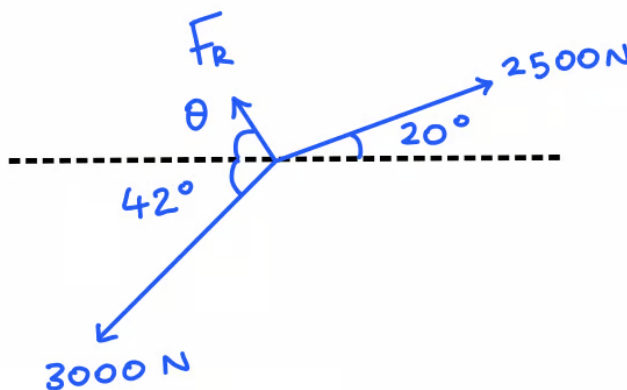
- For a scale of 1 cm : 500 N
- Acceptable range for measurement: 2.2 – 2.5 cm [1 mark]
- Acceptable range for F_R : 1100 – 1250 N [1 mark]

[Total: 4 marks]

Note that the boat in this question is moving at a **constant velocity**: therefore, your vector triangle must be **closed** (i.e., the vectors begin and end at the same point, when added tip-to-tail). This is because there is no resultant force acting on the boat!

The question states "by scale diagram or otherwise" meaning you could obtain the marks by calculation instead, however, as you will see below, it involves a few more steps than the scale diagram route

The free-body diagram for the three forces acting on the boat is:



Resolve the forces parallel to the direction of motion:

- $F_R \cos \theta + 3000 \cos 42 = 2500 \cos 20$
- Hence, $F_R \cos \theta = 2500 \cos 20 - 3000 \cos 42$
- $F_R \cos \theta = 120 \text{ N}$

Resolve the forces perpendicular to the direction of motion:

- $F_R \sin \theta + 2500 \sin 20 = 3000 \sin 42$
- Hence, $F_R \sin \theta = 3000 \sin 42 - 2500 \sin 20$
- $F_R \sin \theta = 1150 \text{ N}$

Determine the angle θ :

- Since $\tan \theta = \frac{\sin \theta}{\cos \theta}$ then $\frac{F_R \sin \theta}{F_R \cos \theta} = \tan \theta = \frac{1150}{120}$

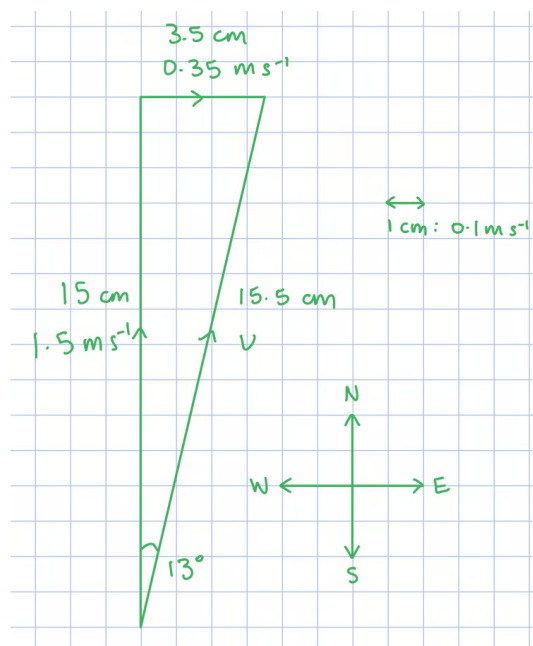
- Hence, $\theta = \tan^{-1}\left(\frac{1150}{120}\right) = 84^\circ$

Use the angle θ to calculate the magnitude of F_R :

- $F_R \sin \theta = 1150 \text{ N}$ and $F_R \cos \theta = 120 \text{ N}$
- Therefore, $F_R = \frac{1150}{\sin 84} = \frac{120}{\cos 84} = 1160 \text{ N}$

3d

(d) Determine the resultant velocity of the boat:



- Scale indicated as $1 \text{ cm} : 0.1 \text{ m s}^{-1}$ (or any other appropriate scale); [1 mark]
- Correct resultant velocity v drawn, with direction indicated (as shown); [1 mark]

Measurement of resultant velocity:

- For a scale of $1 \text{ cm} : 0.1 \text{ m s}^{-1}$
- Acceptable range for measurement: $15 - 16 \text{ cm}$
- Acceptable range for velocity: $1.5 - 1.6 \text{ m s}^{-1}$ [1 mark]
- Acceptable range for direction: $11 - 15^\circ$ east of north / clockwise from north (or equivalently expressed); [1 mark]

[Total: 4 marks]

Challenging questions will not prompt you to include the requisite information for a **resultant velocity**, i.e., both **magnitude and direction**. Remember, this is a vector – hence a **scale construction** needs to precisely identify both components of this

